

# DeepTrace Hackathon Execution Plan

Professional execution roadmap for the Pirates of the Coral-bean Hackathon. Project Focus: Coral-powered Production Commit Intelligence System. Timeline: May 26 to Final Submission Day.

## 1. Project Summary

DeepTrace is a web-based operational intelligence system designed to detect suspicious or production-breaking commits by correlating GitHub repository activity, operational logs, deployment events, and team discussions using Coral as the unified retrieval and orchestration layer. Primary Goals: Detect risky commits and deployment correlations. Analyze uploaded logs and operational evidence. Provide AI-generated root cause analysis and recommendations. Demonstrate meaningful Coral integration through multi-source joins and retrieval workflows.

## 2. Finalized Technology Stack

| Layer              | Technology                     |
|--------------------|--------------------------------|
| Frontend           | Next.js + Tailwind + shadcn/ui |
| Language           | TypeScript                     |
| Data Orchestration | Coral                          |
| AI Reasoning       | Gemini API                     |
| Development AI     | Ollama                         |
| Validation         | Zod                            |
| Version Control    | Git + GitHub                   |
| Deployment         | Vercel                         |

## 3. Git Workflow and Recovery Strategy

Repository Structure: **main** — Stable demoable builds only. **dev** — Active development branch. **feature/\*** — Isolated feature branches.

| Time                  | Required Action                                     |
|-----------------------|-----------------------------------------------------|
| Before Work Session   | Pull latest changes and create recovery checkpoint. |
| Every 2–3 Hours       | Commit and push incremental progress.               |
| Before Major Refactor | Create backup branch and push current stable state. |
| Before Sleeping       | Mandatory stable push + zip backup export.          |

Recovery Rules: Maintain nightly compressed project backups. Tag stable milestones (v0.1, v0.2, etc.). Never leave the project in a broken state overnight. Preserve at least one stable demoable build at all times.

## May 26 — Foundation Day

| Time Block  | Task                                                                                          |
|-------------|-----------------------------------------------------------------------------------------------|
| 09:00–12:00 | Initialize Next.js project, TypeScript strict mode, Tailwind, shadcn/ui, ESLint, and Prettier |

|             |                                                        |
|-------------|--------------------------------------------------------|
| 13:00–16:00 | Finalize folder architecture and project organization. |
| 17:00–20:00 | Install Coral and test basic local retrieval/querying. |
| 20:00–21:00 | Documentation update, commit, push, and backup export. |

**End-of-Day Checkpoints:**

- Project boots successfully.
- Coral installation verified.
- Repository structure finalized.
- Stable initial commit pushed.

## May 27 — Data Layer Integration

| Time Block  | Task                                                                 |
|-------------|----------------------------------------------------------------------|
| 09:00–12:00 | Implement GitHub API integration for commits, PRs, and file changes. |
| 13:00–15:00 | Implement log upload and parsing pipeline.                           |
| 15:00–18:00 | Connect Coral with GitHub and uploaded logs.                         |
| 18:00–20:00 | Buffer window for integration debugging and schema validation.       |
| 20:00–21:00 | Push stable build and create backup.                                 |

**End-of-Day Checkpoints:**

- GitHub integration working.
- Logs upload functioning correctly.
- Coral joins verified.

## May 28 — AI Investigation Workflow

| Time Block  | Task                                            |
|-------------|-------------------------------------------------|
| 09:00–11:00 | Define structured prompts and response schemas. |
| 11:00–14:00 | Integrate Ollama for local AI inference.        |
| 15:00–18:00 | Build first complete investigation workflow.    |
| 18:00–20:00 | Error handling and validation improvements.     |
| 20:00–21:00 | Tag stable milestone and push backup.           |

**End-of-Day Checkpoints:**

- End-to-end investigation flow working.
- AI responses structured and stable.
- First usable demo build created.

## May 29 — UI and Visualization

| Time Block  | Task                                                       |
|-------------|------------------------------------------------------------|
| 09:00–12:00 | Build dashboard interface and query input flow.            |
| 13:00–16:00 | Implement timeline visualization and report cards.         |
| 16:00–18:00 | Add loading states, transitions, and retrieval indicators. |

|             |                                      |
|-------------|--------------------------------------|
| 18:00–20:00 | Responsive layout and UI refinement. |
| 20:00–21:00 | Push stable UI build and backup.     |

**End-of-Day Checkpoints:**

- Professional dashboard completed.
- Timeline visualization functioning.
- UI visually polished and stable.

## May 30 — Intelligence and Stabilization

| Time Block  | Task                                                 |
|-------------|------------------------------------------------------|
| 09:00–12:00 | Improve commit correlation and risk scoring logic.   |
| 13:00–15:00 | Integrate Gemini API for final reasoning quality.    |
| 15:00–17:00 | Minimal Slack integration or Slack export ingestion. |
| 17:00–19:00 | Performance optimization and cleanup.                |
| 19:00       | Hard feature freeze begins.                          |
| 20:00–21:00 | Stable push, tag, and backup.                        |

**End-of-Day Checkpoints:**

- Core functionality complete.
- Feature freeze enforced.
- Stable demo-ready build available.

## May 31 — Final Submission Day

| Time Block    | Task                                                   |
|---------------|--------------------------------------------------------|
| 09:00–12:00   | Bug fixing and stability testing only.                 |
| 13:00–15:00   | Prepare screenshots, README, and architecture diagram. |
| 15:00–17:00   | Record and rehearse demo multiple times.               |
| 17:00–19:00   | Deploy and validate production build.                  |
| 19:00–21:00   | Submission upload and final verification.              |
| Final 2 Hours | No coding. Verification and recovery only.             |

**End-of-Day Checkpoints:**

- Submission assets complete.
- Deployment verified.
- Demo rehearsed and stable.

## 4. Critical Operational Rules

No uncontrolled feature additions after feature freeze. Never merge untested code directly into the main branch. Keep all modules typed and documented. Preserve a working demo state at all times. Prioritize stability and clarity over feature quantity. Do not rewrite architecture during the hackathon. Avoid large unreviewed AI-generated code dumps.

This execution plan is designed to optimize reliability, maintain clean engineering practices, reduce integration risk, and maximize the probability of delivering a polished and stable hackathon submission.